

SHUBHASHIS SINHA

Senior Staff AI Engineer — Enterprise AI Platform · RAG & Agents · MCP / MCP Hub · Bedrock AgentCore ·
Microservices/APIs

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SUMMARY

Senior Staff AI engineer with **12+ years** leading **enterprise full-stack and AI platform development at scale** across multiple initiatives. I design and ship **microservices and REST APIs**, build **RAG, multimodal LLM features, and agents**, and set standards for **AI-assisted development** — owning technical decisions end-to-end while balancing reliability, performance, and cloud cost. Proven track record shipping production **LLM, RAG, and MCP** systems at scale, mentoring engineers, guiding contractors and partners, and driving Agile delivery with prompt/model testing and observability. Currently Staff AI Architect at GE HealthCare's AI Center of Excellence.

CORE COMPETENCIES

AI Platform & Services	Enterprise AI Platform (at scale) · AI Reusable Services · RAG & Multimodal LLM Features · Agents & Agentic Workflows · MCP & MCP Hub (tool registry) · Amazon Bedrock AgentCore (runtime, memory, gateway, identity) · Reliability / Performance / Cloud-Cost Optimization
Microservices & APIs	Resilient REST APIs · Microservices · Event-Driven Architecture · FastAPI · Service Design & Contracts · Full-Stack Delivery · Front-End (JavaScript/TypeScript, React — AI-assisted)
LLM & Retrieval	Enterprise RAG Architectures · Semantic Search Optimization · Conversation Memory at Scale · Embeddings & Vector DBs (pgvector, FAISS, OpenSearch; Milvus/Pinecone/Weaviate-mappable) · Multiple LLM Providers · LangChain / LlamaIndex (aware)
AI-Assisted Development	Standards for AI-Assisted Dev · Agentic Coding Tools (Claude Code / Kiro, daily) · Prompt & Model-Performance Testing · Code-Quality Automation · Unit Testing
Cloud & Infra	AWS (Bedrock, EKS, Lambda, S3, OpenSearch, SageMaker, Step Functions) · Docker · Kubernetes · CI/CD · Observability for AI Features (LangSmith; token/latency/cost) · Python (preferred)
Leadership	Technical Leadership · Mentoring Engineers · Guiding Contractors & Vendors · Code Reviews & Integrations · QA Methodologies · Agile · Measurable Impact

PROFESSIONAL EXPERIENCE

Staff AI Architect — AI Center of Excellence (GE experience — 4+ years)

Dec 2021 – Present

GE HealthCare · Bangalore, India

Enterprise AI Platform — RAG, Multimodal LLM & Agents

- Led **enterprise AI platform development at scale** across multiple initiatives — building **RAG and multimodal LLM features and agents**, and identifying/delivering **reusable AI services** adopted across teams.
- Designed **agentic workflows for autonomous tasks** on LangGraph with **MCP** tool interfaces — a GraphRAG multi-agent platform and a Contract Digitalization system processing **1,000+ documents/month**.
- Standardized agent tooling via an **MCP Hub** — a centralized registry of reusable MCP servers/tools shared across agents and teams — and deployed/scaled agentic workloads on **Amazon Bedrock AgentCore** (runtime, memory, gateway, identity) for secure, managed enterprise deployment.
- Built **enterprise RAG architectures** with semantic-search optimization and conversation memory over 10,000+ documents at sub-second P95.

Microservices, APIs & Full-Stack Delivery

- Designed and shipped **resilient REST APIs and microservices** (FastAPI) powering AI applications and services, with event-driven integration.
- Delivered **full-stack AI features** — backend AI services integrated with front-end interfaces (JavaScript/TypeScript, React; AI-assisted) and rapid Streamlit UIs.
- Owned **technical decisions end-to-end**, balancing reliability, performance, and cloud cost for AI systems.

AI-Assisted Development, Testing & Observability

- Set **standards for AI-assisted development** — daily use of agentic coding tools (Claude Code / Kiro) to accelerate build, test, and review across teams.
- Established **testing for prompts and model performance** (RAGAS, LLM-as-Judge; validator agent ~95% correctness) and **code-quality automation** with unit testing.
- Implemented **observability for AI features** — LangSmith tracing, token/latency/cost monitoring, and drift detection in production.

Technical Leadership & Agile Delivery

- Provided **technical leadership**, mentored **8+ engineers**, and guided external contractors and partners; ran code reviews and integrations.

- Drove **Agile practices** and a reusable framework adopted by 6+ teams (25% faster development).
- Delivered measurable impact — platform migration (\$2M+ savings) and AI automation reducing effort ~40%.

Tech Stack: Python · LangGraph · MCP / MCP Hub · Amazon Bedrock AgentCore · RAG / Multimodal LLM · Agents · Vector DBs (pgvector, FAISS, OpenSearch) · FastAPI / REST / Microservices · JavaScript/TypeScript · React (AI-assisted) · AWS (Bedrock, EKS, Lambda, S3, OpenSearch, SageMaker) · Docker · Kubernetes · CI/CD · RAGAS / LLM-as-Judge · LangSmith

Senior Software Engineer — Backend & Distributed Systems

Oct 2017 – Dec 2021

Bank of America · Chennai, India

- Designed and shipped **production-grade microservices, REST APIs, and distributed systems** — processing **5TB+ daily** — with reliability, performance, and cost discipline.
- Optimized **complex processing** — reduced query execution time 40% on 100B+ row datasets.
- Built **automated testing and code-quality controls** (reduced manual QA 30%) with CI/CD and code review.
- Mentored engineers and led integrations across teams in an Agile environment.

Tech Stack: Python, Scala, SQL, Apache Spark, Snowflake, Kafka, AWS (S3, EMR, Glue), Kubernetes, CI/CD

Software / Data Engineer

Mar 2014 – Oct 2017

T-Mobile · Bangalore, India

- Built **large-scale data pipelines and services** (Hadoop, Pig, Hive) across **5M+ subscribers** — 25% improvement in data-driven decision making.
- Implemented **CDC/incremental ingestion** and optimized query performance 30% — near real-time data freshness.

Tech Stack: Hadoop (CDH), Apache Pig, Hive, Sqoop, Python, SQL

EDUCATION

Bachelor of Technology — Electronics & Communications Engineering

West Bengal University of Technology · Kolkata, India · GPA: 8.87/10

AWARDS & CERTIFICATIONS

 **Sir M. Visvesvaraya Award — GE HealthCare (2024)**
Recognized for shipping agentic/GenAI systems with measurable production impact serving real users.

 **Impact Award — GE HealthCare (Oct 2022, Nov 2023, Mar 2024)**
Reusable framework development, platform migration delivering \$2M+ savings, and AI-driven automation.

 **AWS Certified Solutions Architect – Associate (2025) · Patent filing in progress (GraphRAG SPARQL Validator architecture)**